

Linear growth under an unbounded additive driver

ziangni-sys

Conjecture 00000008480

The source defines nonautonomous optimization through the averages $n^{-1} \sum_i (f(T^i x) + h_i)$ and asserts sublinear growth of their supremum under unbounded driving. Here the supremum is over the initial state x for the displayed finite averaging window. We disprove this assertion using the unbounded driver $h_i = i$.

Theorem 1. *There is a compact probability-preserving ergodic system with a continuous observable and an unbounded nonnegative driver such that the optimized averages grow linearly, rather than sublinearly.*

Proof. Let $\Omega = \{*\}$ with its usual topology and full sigma-algebra, let $\mu = \delta_*$, and let T be the identity. This space is compact, T is continuous and preserves probability, and it is ergodic: its only measurable sets are $\emptyset$ and Ω . Set $f(*) = 0$, a continuous observable, and $h_i = i$ for every integer $i \geq 0$. This driver is nonnegative and unbounded, since for every $B \in \mathbb{R}$ there is an integer $i > B$.

For a fixed window origin $k \geq 0$ and $n \geq 1$, define

$$A_{k,n}(x) = \frac{1}{n} \sum_{i=0}^{n-1} (f(T^i x) + h_{k+i}), \quad M_{k,n} = \sup_{x \in \Omega} A_{k,n}(x).$$

The source's displayed average is the case $k = 0$. Since Ω has one element, the set being optimized is the singleton $\{A_{k,n}(*)\}$; its supremum is attained and equals that number. The arithmetic sum gives

$$M_{k,n} = \frac{1}{n} \sum_{i=0}^{n-1} (k+i) = \frac{nk + n(n-1)/2}{n} = k + \frac{n-1}{2}.$$

Thus, for every fixed k ,

$$\frac{M_{k,n}}{n} = \frac{1}{2} + \left(k - \frac{1}{2}\right) \frac{1}{n} \longrightarrow \frac{1}{2} \neq 0.$$

In particular, $M_{0,n} = (n-1)/2$ is not $o(n)$. The claimed universal sublinear growth under unbounded driving therefore fails. $\square$

Scope

Both the English and Chinese versions state the unbounded-driver clause without a growth or cancellation restriction on the driver. Our driver is as simple as a nonnegative linear sequence; the autonomous observable is bounded and identically zero. The example satisfies the usual compactness, continuity, probability preservation and ergodicity conditions as well. The proof refutes the sublinear clause, so no claim about the other bounded or random-driver clauses is required.

Which supremum is computed

The optimized set here consists of values over initial states for a given finite n -term average, exactly as specified by the displayed expression in the source. The extra parameter k shows that the counterexample works at every fixed starting time. It is held fixed while n tends to infinity. We do not identify this finite-window supremum with a different operation that also takes a supremum over all starting times. Such an additional operation is not needed for the counterexample.

Formal correspondence

The Lean project defines the actual one-point Dirac probability measure and identity map. It proves `CompactSpace`, continuity of both T and f , and Mathlib’s actual `Ergodic` predicate, which includes measure preservation. The theorem `driver_unbounded` states $\forall B \in \mathbb{R}, \exists i \in \mathbb{N}, B < h_i$.

The definition `average` uses an actual finite sum over `Finset.range n`, actual iterates T^i , the observable, and the shifted driver. The definition `optimal` is the actual real supremum `sSup (Set.range (average start n))`. The complete image set is proved to be a singleton in `average_range`; `optimal_attained` then uses the genuine singleton supremum theorem. Thus no scalar formula for the supremum is assumed.

The arithmetic identity is proved by induction on the natural index in `sum_formula`. The theorem `shifted_sum` accounts for every term of the shifted window, and `optimal_formula` establishes

$$\forall k, n \in \mathbb{N}, \quad n \geq 1 \implies M_{k,n} = k + (n - 1)/2.$$

The real reciprocal limit gives `ratio_tendsto_half`. Finally, `not_sublinear` proves, for every fixed natural k , the negation of Mathlib’s actual `Asymptotics.IsLittleO` predicate for `optimal k` relative to $n \mapsto n$ at infinity. If that predicate held, the ratio would tend to zero, contradicting uniqueness of its proved limit $1/2$.

The index zero is harmless: Lean uses totalized division there, whereas every displayed finite-window formula requires $n \geq 1$. All convergence claims are at infinity, and are unaffected by that single initial index.

Reproduction

The project pins Lean 4.19.0 and Mathlib commit

```
c44e0c8ee63ca166450922a373c7409c5d26b00b.
```

From `lean/`, run `lake update`, `lake exe cache get`, and `lake build`. Check the complete source with

```
lake env lean -DwarningAsError=true Main.lean.
```

Six printed dependency audits cover the system, driver, actual supremum, formula, limit and final negation. Only standard logical axioms are used. No admitted statements, extra axioms or auxiliary numerical computations are needed.